

3	(a)	(i)	(constant speed, so) acceleration (is always) <u>normal / perpendicular to velocity</u> <u>velocity tangent to circle</u> (at any point), so acceleration is directed towards centre	B1 B1
		(ii)	$a = \frac{v^2}{r}$	B1
	(b)	(i)	force on rotating blades $= \frac{\Delta m}{\Delta t} \times v$ $\frac{\Delta m}{\Delta t} = \rho(\pi r^2)v = 1.2 \times \pi \times 0.6^2 \times v$ force on blades equal weight when hovering $1.2 \times \pi \times 0.6^2 \times v \times v = 1.8 \times 9.81$ $v = 3.61 \text{ m s}^{-1}$	C1 M1 A1
		(ii)	1. provide sufficient upward force in lower air density by having a higher <u>air</u> speed, so as to move larger mass (or volume) of air <u>per unit time</u>	B1
			2. higher angular velocity so higher linear speed, <u>centripetal acceleration higher</u> (at every point on blades) centripetal acceleration (or force) <u>provided by tension / tensional force in blades</u> so, larger stress in the blades	B1 B1

4	(a)	(i)	1	constant phase difference (and does not vary with time) between the waves	B1
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